

Course Structure



Dr. Vishwanath Karad

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TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

Course Structure					
Course Code	BSC10050				
Course Category	PF				
Course Title	Microprocessor				
Teaching Scheme	Lectures	Tutorials	Laboratory / Practical	Project	Total
Weekly load hours	3		2		5
Credits	3		1		4
Assessment Code	TL3				

Assessment Structure



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Type of Course	Assessment Scheme Code	Description L-T-P-J-C	CCA1	MT	CCA2	LCA1	LCA2	LCA3	TE	Total
Theory and Lab Course 3	TL3	3-0-1-0-4	10	25	10	5	5	5	40	100

Course Outcomes



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Course Outcomes:

On completion of the course, student will be able to–

1. Understand the block diagram of 8086 Microprocessor
2. Understand the Instruction set of 8086
3. Write an assembly language program
4. Understand the advanced Microprocessor

Course Contents



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Course Contents:

Unit 1: 8086 Architecture

10 Hours

Evolution of Microprocessors and types, 8086 Microprocessor, Salient features, Pin descriptions, Architecture of 8086 - Functional Block diagram, Register organization, Concepts of pipelining (5 stage), Memory segmentation, Physical memory addresses generation., Minimum mode and Maximum Mode of 8086

Unit 2: Instruction Set of 8086 Microprocessor

15 Hours

Addressing modes, Data transfer instructions, Arithmetic Instructions, Logical Instructions, Bit manipulation instructions, String Operation Instructions, Program control transfer or branching Instructions, Process control Instructions, examples based on Instruction Set

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Unit 3: 8086 Assembly Language Programming

10 Hours

Introduction to Assembly Programming and Assembler Directive

Assembly program on Addition, Subtraction, Multiplication and Division

Sum of Series

Smallest and Largest numbers from array

Sorting numbers in Ascending and Descending order

Block transfer

String Operations - Length, Reverse, Compare, Concatenation, Copy

Unit 4: Advanced Processor

10 Hours

Introduction to Pentium Processor, Feature of Pentium Pro Processor Architecture of Pentium Pro Processor,

Software Model of Pentium Pro , Pipelining of Pentium Processor , Classification of Advanced Processors

Course Contents - Lab



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Lab Experiments for Microprocessor

Assembly language program for arithmetic operations

Assembly language program to convert the given decimal number to binary, octal number system

Assembly language program to find the largest and smallest among the given numbers

Assembly language program to arrange the given numbers in ascending and descending order

Assembly language program to find the square of a number

Assembly language program to find sum of an array

Resources



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Learning Resources:

Reference Books:

- 1) Microprocessor & interfacing (programming & hardware) Revised Second Edition By Douglas V. Hall
Tata McGraw Hill
- 2) Microprocessor Architecture By B.Ram
- 3) Microprocessor Architecture, Interfacing & Programming By Mohammad Raffiqzamman

Supplementary Reading:

- 1) The Intel Microprocessors : Barry B. Brey- Pearson Education Asia
- 2) The Pentium Microprocessor : James Antonakos

Content

- ✓ Introduction to Pentium Processor
- ✓ Feature of Pentium Pro Processor
- ✓ Architecture of Pentium Pro Processor
- ✓ Software Model of Pentium Pro
- ✓ Pipelining of Pentium Processor
- ✓ Classification of Advanced Processors



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Introduction to Pentium Processor



- ✓ It is CISC microprocessor and introduced in 1993.
- ✓ It is packed in a 273-Pin Grid Array (PGA) package.
- ✓ It uses BiCMOS process technology. It is combination of Bipolar junction transistor and CMOS technology.
- ✓ Pentium processor was available in 66MHz to 99 MHz clock frequency.
- ✓ Due to high clock frequency, the processor can execute the instructions at higher rate of speed.
- ✓ It has superscalar architecture like it has three execution unit.
- ✓ One executes floating-point instructions with 8 stage pipelining, and the other two (U-pipe and V-pipe) execute instructions with five stage pipelining.
- ✓ It has 64 bit data bus and memory is byte addressable so it require 8 level memory banking.
- ✓ It has 32 bit address bus.

Introduction to Pentium Processor



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- ✓ It can support 64TB virtual memory with the help of segmentation and paging techniques.
- ✓ It support Integer UV Pipelining.
- ✓ It supports branch prediction algorithm with 95% accuracy so that processor doesn't require to fetch instructions and predict well in advance.
- ✓ It has on chip L1 Cache memory. The cache has been reorganized to form two caches that are each 8K bytes in size, one for caching data, and the other for instructions.
- ✓ Due to on chip cache, there is no delay as happened in earlier microprocessor due to off chip cache memory because of bus interface.
- ✓ Pentium processor employs even parity checking for data integrity, ensuring that errors are detected promptly.

Architecture of Pentium Processor



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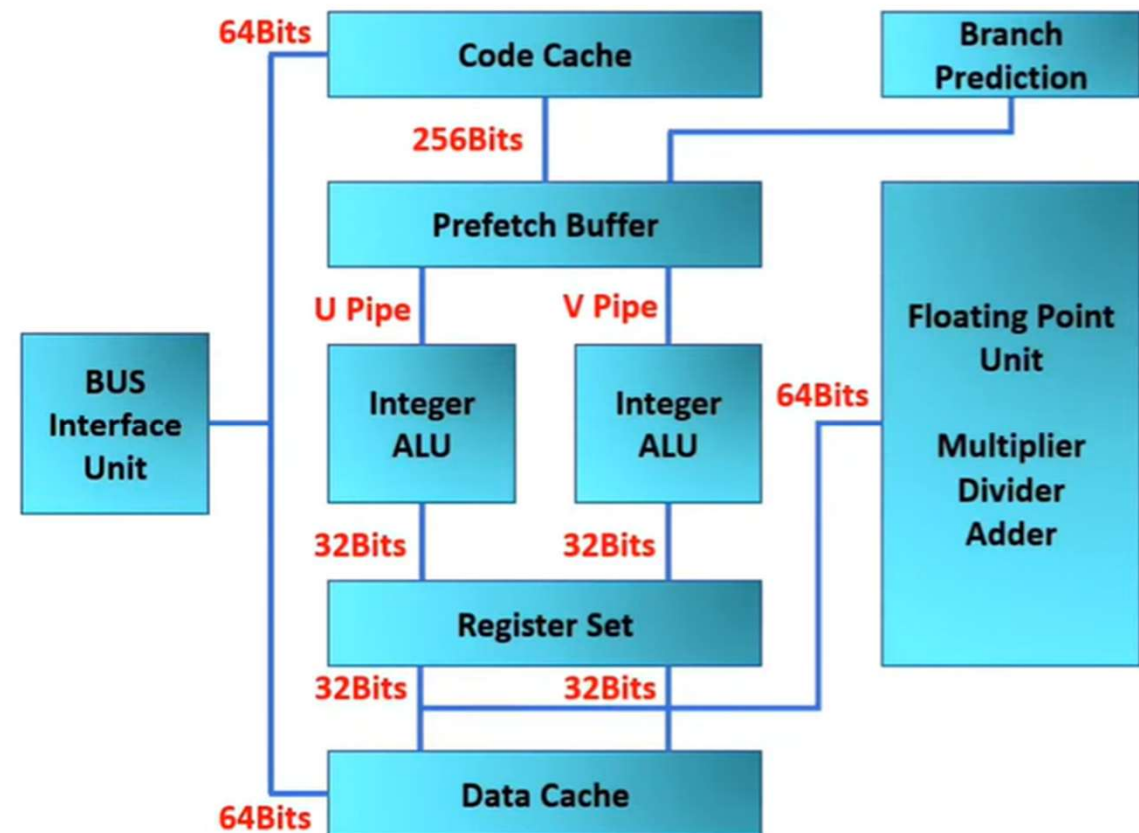
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❖ BUS Interface Unit:

- ❑ BIU controls all the system bus of MPU.
- ❑ Addressing of Memory and IO is done by BIU.
- ❑ Pentium has 64 Data bus and 32 Address bus.

❖ Superscalar Structure:

- ❑ Superscalar means this architecture has more than one execution units.
- ❑ Pentium processor contains two integer units:
 - U Pipe
 - V Pipe
- ❑ Each of this integer unit has 32bits of ALU.
- ❑ Due to two execution unit, performance of Pentium also doubled.



Architecture of Pentium Processor



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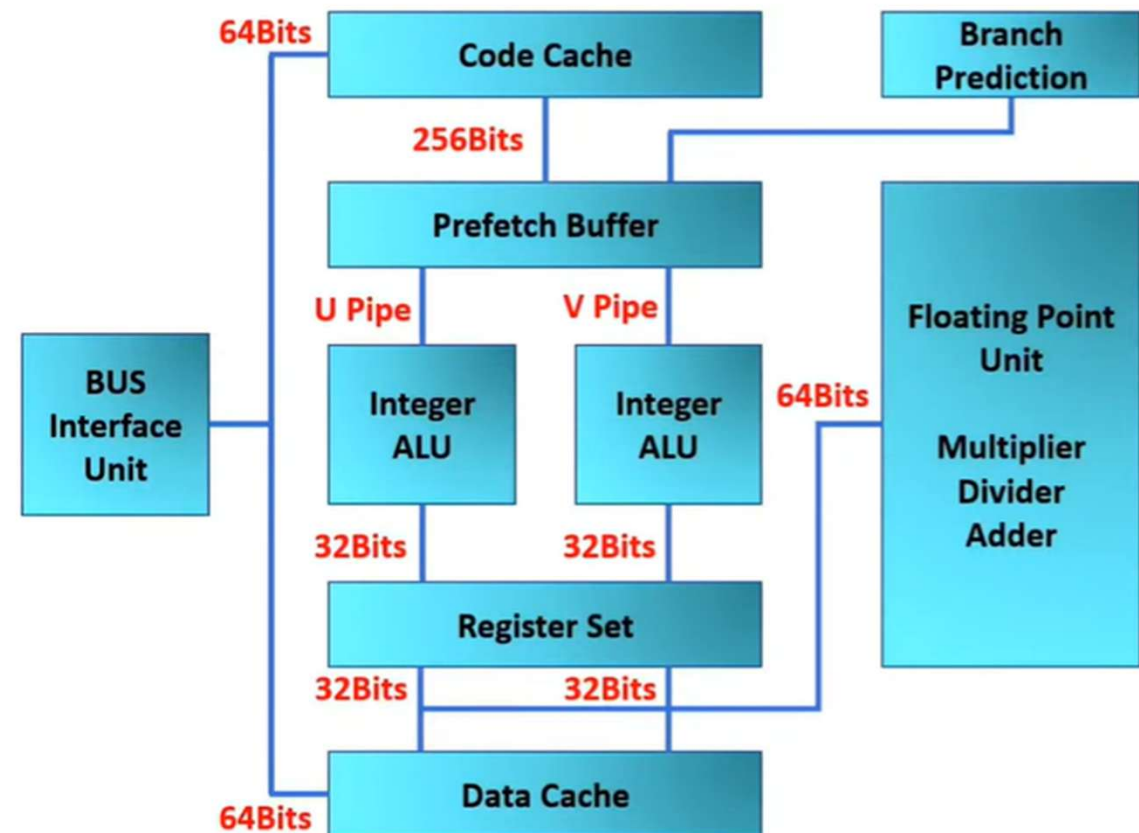
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❖ Prefetch Unit:

- ❑ Prefetch unit is used to implement FIVE stage pipelining.
- ❑ It consists pair of prefetch buffers having a size of 32Bytes each.
- ❑ Both buffers operate independently but not at the same time.
- ❑ It takes instructions sequentially from Code Cache.

❖ Branch Prediction Unit:

- ❑ Branch Prediction plays essential role in the prediction of branch.
- ❑ Branch instruction causes pipeline stalling.
- ❑ Frequent pipeline stalling degrades performance of execution.
- ❑ So by branch prediction we can predict branch instruction in advance.
- ❑ Branch prediction is done with the use of BTB Branch Target Buffer stores 256 entries.



Architecture of Pentium Processor



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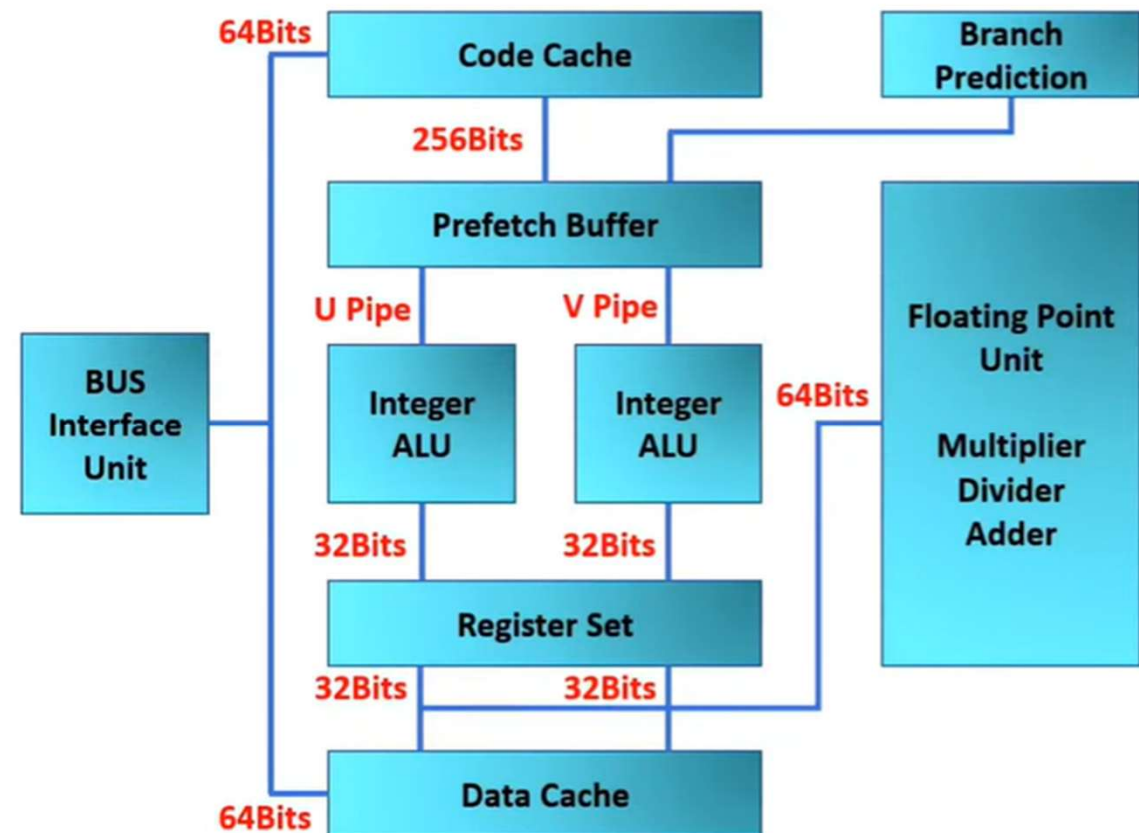
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❖ Execution Unit:

- ❑ Two execution units are executing instructions in parallel.
- ❑ Each execution unit has 32bits of ALU.
- ❑ As per the execution of instruction, it updates the data of register set.

❖ Cache Memory:

- ❑ Pentium has integrated cache memory, which increases the execution speed of processor.
- ❑ It has 16Kb split Cache memory integrated on chip for data and for Code.
- ❑ Split Cache reduces the memory conflicts.



Architecture of Pentium Processor



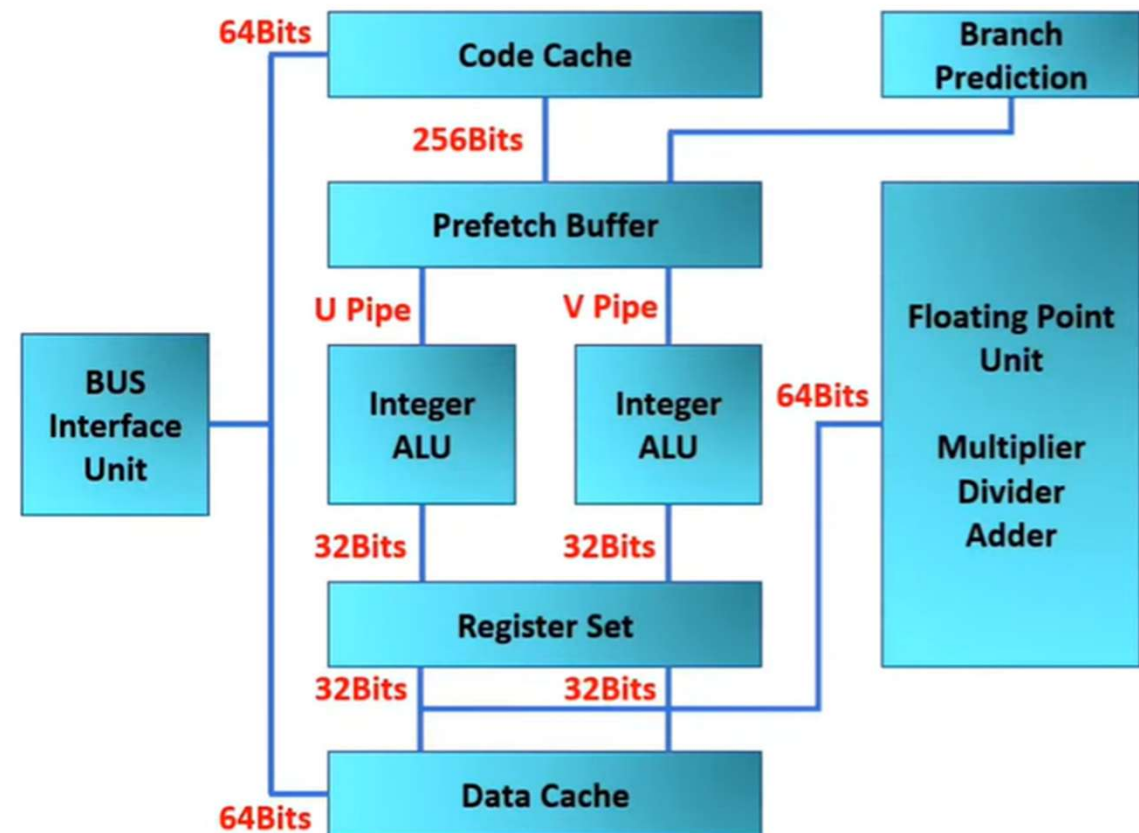
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❖ Floating Point Unit:

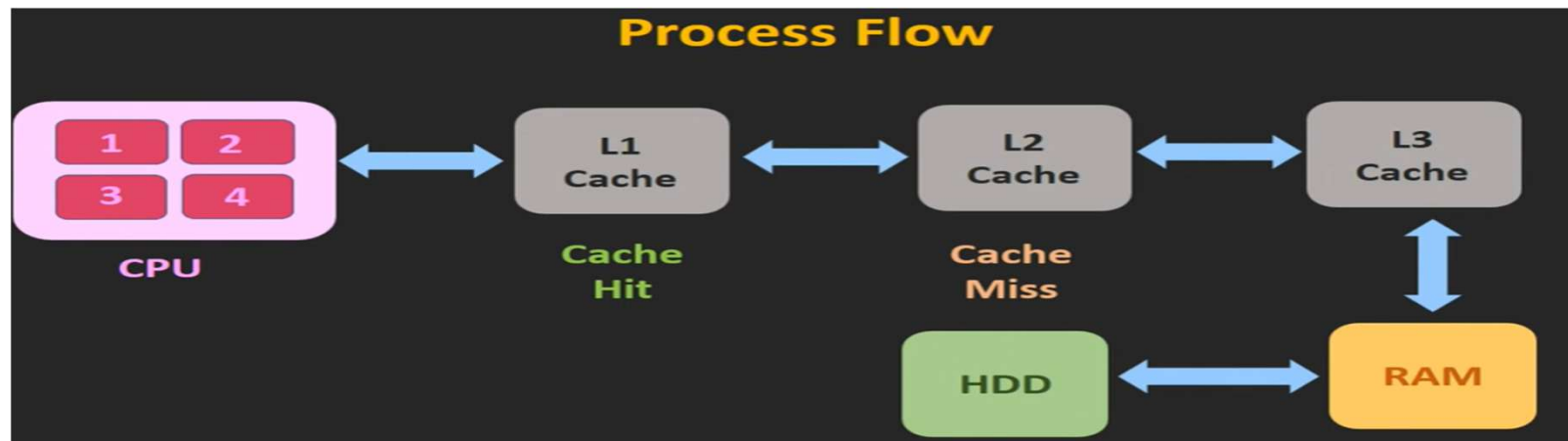
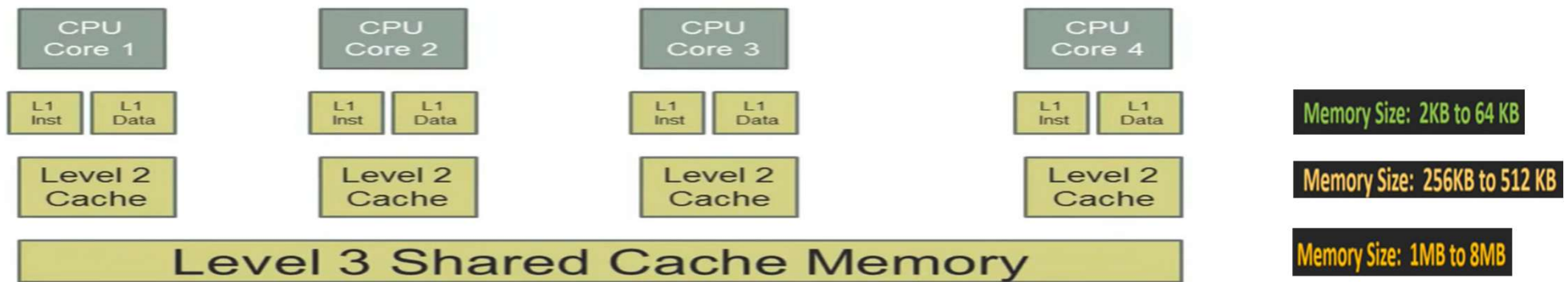
- ❑ It is used for execution of floating point instructions.
- ❑ Floating point unit also supports 8 stage pipeline.
- ❑ Floating point unit enhances the capabilities of microprocessor for graphics & multimedia applications.
- ❑ It has dedicated hardwire circuits for Multiplier, Divider & Adder.
- ❑ The hardwire control improves performance.
- ❑ FPU supports single, double & extended precision floating point operations with eight 80 bits registers.



Cache Memory Level



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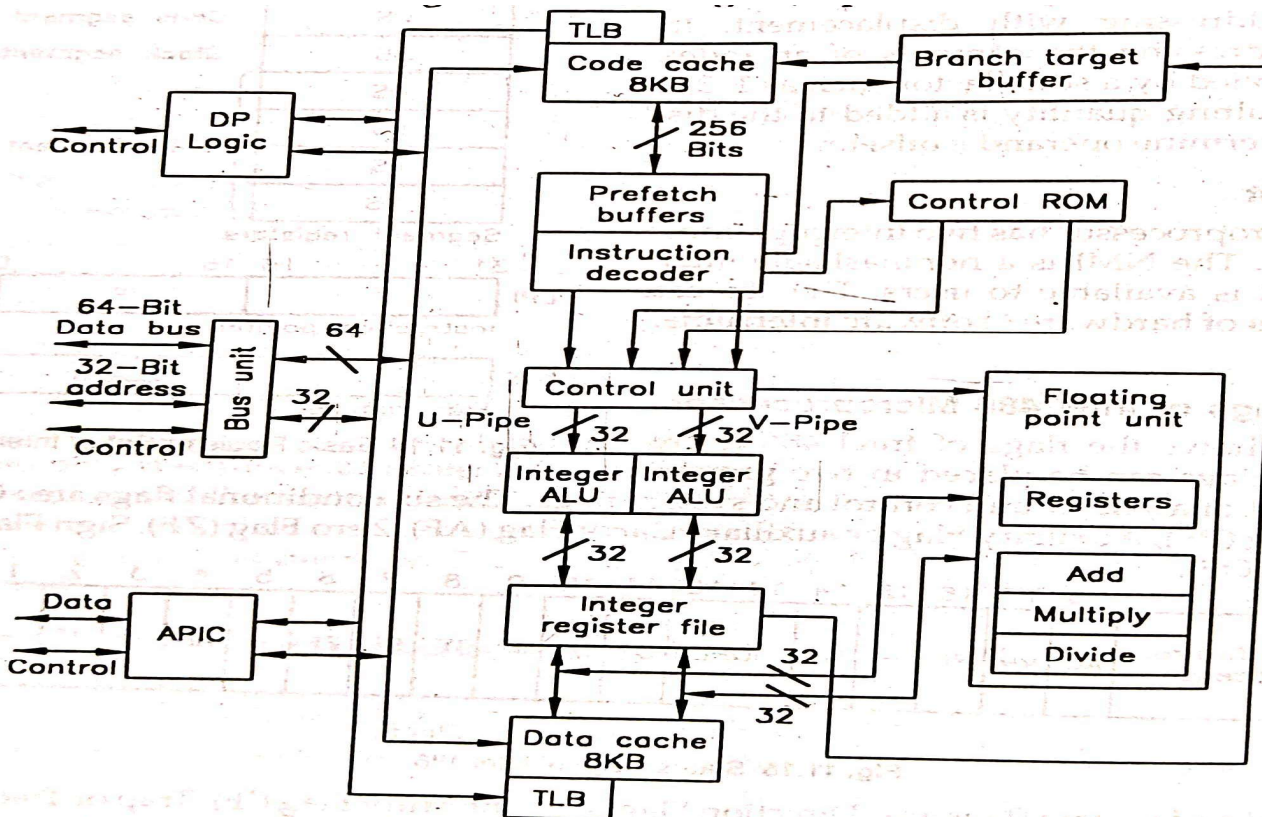
Pentium Processor – Block Diagram



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Block Diagram of Intel Pentium Processor

Pentium Processor – Block Diagram



- ✓ **Bus Interface Unit** – BIU controls all bus system in processor. It controls the addressing of memory and input/output units of the system. It has 32 bit address bus and 64 bit data bus.
- ✓ **Superscalar Structure** – It means that Pentium has more than one execution units. It has two integer units (32 bit U-Pipe and 32 bit V-Pipe) and one 64 bit Floating point unit. Due to two integer unit, the performance has increased to double fold.
- ✓ **Prefetch Unit** – It is used for pipelining operation with five stage and we have two pile i.e. U and V pile. So prefetch unit has two prefetch buffer for each pipe. Both buffers operate individually with 32 bytes and not at the same time. It takes instructions sequentially from code cache.
- ✓ **Branch Prediction Unit** – It is used to improve the performance of the system because it prevents the flushing the pipeline due to branch instruction. It is done with the use of Branch Target Buffer (BTB) stores 4 way set associative mapping of 256 entries. Frequent pipeline stalling degrades the performance of execution.
- ✓ **Execution Unit** – It has two execution units with U-pipe and V-pipe in parallel. So these can execute 10 instruction at a time. Each unit has 32 bits of ALU. As per the execution of instruction, it updates the data in register set.

Pentium Processor – Block Diagram



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- ✓ **Cache Memory Unit** – It has integrated cache memory which increases the execution speed of processor. It is on-chip cache memory which remove delay due to bus integration. It is splited into two parts as 8 KB of code cache and 8KB of data cache.
- ✓ **Floating Point Unit** – In the earlier versions of processors, floating points operations are done with the help of math coprocessors but in Pentium processors, it has separate floating point unit. It has 8 stage pipeline. It enhances the capabilities of microprocessor for graphics and multimedia applications. It has dedicated hardwired circuits for multiplier, divider and adder. The hardwired unit improves the performance a lot.

Pentium Vs Pentium Pro Processor



Pentium Processor	Pentium-Pro Processor
The Pentium processor is the fifth-generation processor 80586 and it is represented by P5. This processor was developed in 1993 by Intel.	The Pentium-Pro processor is the sixth-generation processor and it is represented by the P6 family processors. This processor was developed in 1995 by Intel.
This is a 5 V processor and it was fabricated in 0.8-micron BiCMOS technology	Pentium-Pro processor core operates at 3.1 V to 3.3 V and it is fabricated in a 0.6 μm BiCMOS process.
Pentium runs at a clock frequency of 60 MHz or 66 MHz.	The Pentium-Pro processor's clock speeds are 150, 166, 180 or 200 MHz with a 60 or 66 MHz external bus clock.
Pentium has 3.1 million transistors.	Pentium-Pro consists of 5.5 million transistors.
Pentium has 2 x 8 Kbyte L1 cache, but there is no L2 cache.	Pentium-Pro has 2 x 8 Kbyte L1 cache, and a 256 Kbyte L2 cache.
Pentium has a 32-bit address bus.	Pentium-Pro has a 36-bit address bus.
It has two instruction units.	It has six instruction units.
It has superscalar pipeline. It has two independent integer pipelines and a floating point pipeline.	The Pentium-Pro has a 12-stage decoupled super pipeline architecture which uses an instruction pool. It has speculative execution through register renaming with 40 registers.

Architecture of Pentium Pro Processor

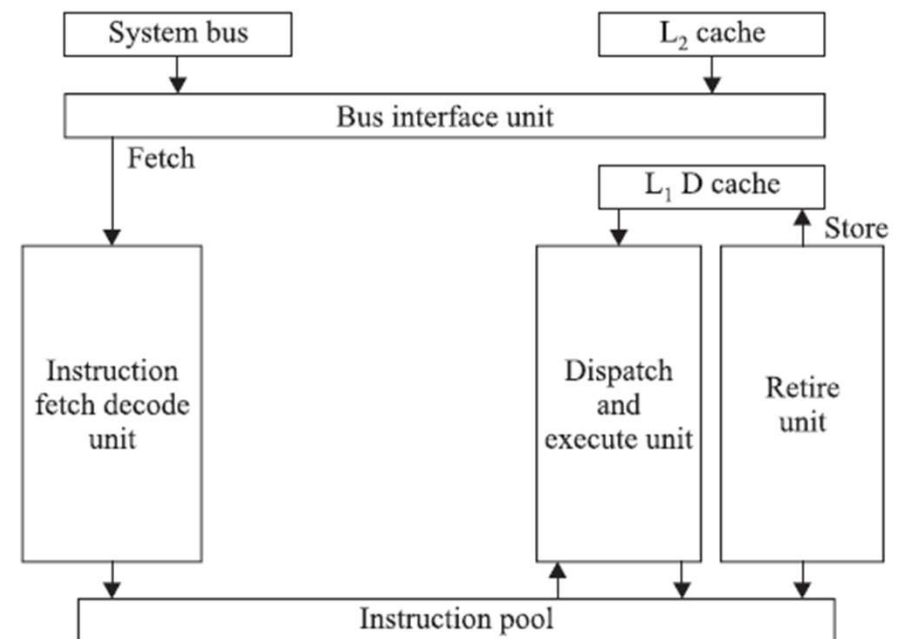


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1. **Fetch/Decode Unit** : Fetch and decode unit receive the program instructions from the instruction cache and decode them into a series of μ operations (μ ops).
2. **Dispatch/Execute Unit** : The dispatch/execute unit accepts the data flow stream, schedules execution of the μ -ops depending on data dependencies and availability of resource and temporarily stores the results of these approximate executions. The dispatch unit selects μ -ops from the instruction pool depending upon their status.
3. **Retire Unit** : The retire unit is the unit that places the temporary or approximate results to permanent locations. This unit knows how and when to retire the temporary, approximate results to permanent architectural state. The retire unit not only checks which μ -ops are complete, it also re-impose the original program order on them.



Architecture of Pentium Pro Processor

Software Model of Pentium Pro Processor



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1. **Memory:** Pentium Pro microprocessor have 8-bit memory. Two consecutive bytes form a word, four consecutive bytes form a double word. All the address in the memory are of byte formats. Programmers usually view Pentium Pro (x86) memory as a collection of segments.
2. **Register:** There are 8 general-purpose registers present in the Pentium Pro architecture. Each register is 32-bit long. First four register are used for data manipulation and next four register are used to hold address. There are some special-purpose registers in the x86 architecture such as Segment register, FLAGS register and IP register.
3. **Data Formats:**
 - Integers are stored as 8-, 16-, 32-bit Binary numbers.
 - Characters are represented using 8-bit ASCII codes.
 - Floating points are represented using three different formats namely single-precision format, double-precision format, extended-precision format.

Software Model of Pentium Pro Processor



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4. Instruction Formats: All the Pentium Pro instruction uses basic format. The basic format for Pentium Pro Instructions is:

- **Prefix:** A Prefix specifies the operation of an instruction.
- **Suffix:** A Suffix specifies the data type of the operands.
- **Modifier:** A modifier specifies the number of operand involved.

The Opcode is the only element that is present in every instruction. Other elements may or may not present or may be of different lengths depending on the operations of instructions.

5. Addressing Modes: Pentium Pro Architecture have a very large number of Addressing modes. Operands value is specified either by using Immediate mode or by using register mode. Operands stored in memory are specified using variation of the Target address (TA) calculation:

$$TA = (\text{base register}) + (\text{index register}) * (\text{scale factor}) + \text{displacement}$$

- **Base register:** Any general-purpose register may be used as a base register.
- **Index register:** Any general-purpose register except ESP can be used as an index register.
- **Scale Factor:** Scale factor may have values 1, 2, 4 or 8.
- **Displacement:** Displacement may have 8-, 18-, or 32-bit values.

Software Model of Pentium Pro Processor



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6. Instruction Set: Pentium Pro Architecture has a large and complex instruction set having more than 400 different machine instructions. An instruction may have zero, one, two, or three operands. There are Register-to-Register instructions, Register-to-Memory instructions and Memory-to-Memory instructions. In Pentium Pro Architecture there are some special-purpose registers to perform operations required in the high-level programming languages.

7. Input and Output: Input instruction transfer one byte, word or doubleword at a time from an I/O port into register EAX. Output instruction transfer one byte, word or doubleword from EAX to an I/O port. Entire string can be transferred using single operation.

Pipelining of Pentium Processor



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Pentium uses a 5-stage pipeline with the following stages in the pipeline-

Prefetch (PF) In the prefetch stage of integer pipeline of the Pentium processor, instructions are fetched from the instruction cache as instructions are stored initially in the instruction cache. After fetching, the CPU aligns the codes properly. As the instructions are of variable lengths, the initial opcode bytes of each instruction must be properly aligned. After completion of the prefetch stage, the decode stages D1 and D2 will be executed.

Decode-1 (D1) In the decode-1 (D1) pipeline stage, the CPU decodes the instruction and generates a control word. The D1 pipeline stage has two parallel instruction decoders. These implement the pairing rules. Only a single control word may be sufficient to start execution of the data transfer, arithmetic and logical operations in RISC processor. This processor supports complex CISC instructions with the help of microcoded control sequencing.

Decode-2 (D2) The decode-2(D2) pipeline stage is required whenever the control word from D1 stage is decoded to complete the instruction decoding. In this stage, the CPU generates addresses for data memory.

Pipelining of Pentium Processor



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Execute (E) The execution stage is used for both ALU operations and data cache access. The data cache is used for data operands and ALU performs arithmetic logic computations or floating-point operations. In the execution stage, all U-pipe and V-pipe instructions, except conditional branches, are verified for correct branch prediction. A microcode is designed to use both the U and V pipes. Therefore, microcode instructions are executed faster on the Pentium than on the 80486.

Write Back (WB) The final stage of the five-stage pipeline is Write Back (WB). In the WB stage, the CPU updates the contents of registers and status of the flag register after completion of execution. In this stage, the V-pipeline conditional branch instructions are verified for correct branch prediction.